

Single-SPAD Implementation of Quantum Key Distribution

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ABSTRACT

A cost-effective implementation of BB84 protocol for QKD, based on the use of a Faraday rotator variable over four states and a single SPAD, is presented.

Keywords: quantum-key distribution, single-photon avalanche photodetector, polarization, Faraday effect.

1. INTRODUCTION

Nowadays encryption is exploited for protecting information exchange in several applications. Nevertheless the security of commonly used algorithms of encryption is based on the extremely high computational cost required for message decryption. On the other hand, quantum key distribution (QKD) allows for exchanging between two users (Alice and Bob) a key in a way which has been proved as unconditional secure, thanks to the fundamental principles of quantum physics. However to make the QKD a reliable and effective widespread solution, it is essential to reduce the cost and expand the scalability.

In the present work we propose a novel implementation of the BB84 protocol, the first example of QKD proposed in 1984 by C. H. Bennett and G. Brassard [1] and still adopted today [2]. In order to reduce the cost, we investigate the feasibility of QKD using only one single-photon avalanche detector (SPAD), placed after the cascade of a Faraday rotator and a polarization analyzer for the measurement of the quantum state of the photon, differently from the standard implementation that employs two SPADs.

2. SINGLE-SPAD QKD IMPLEMENTATION

A common example of BB84 realization is based on the transmission by Alice of a sequence of single photons with controlled states of polarization, as schematized in Fig. 1. For each single photon Alice chooses one of two non-orthogonal polarization bases (either "rectilinear" or "diagonal") and then one of the two states in the given basis. Bob can choose the basis for the single-photon polarization measure by means of a variable Faraday rotator with polarization rotation set to either 0° (rectilinear basis) or 45° (diagonal basis). Then Bob perform the quantum measure of the photon polarization by means of two SPADs at the different output ports of a polarization beam splitter (PBS). Finally, a sifted key bit 1 or 0 is obtained when Alice and Bob choose the same basis and there is correspondingly a photon detection in SPAD 1 or SPAD 2.

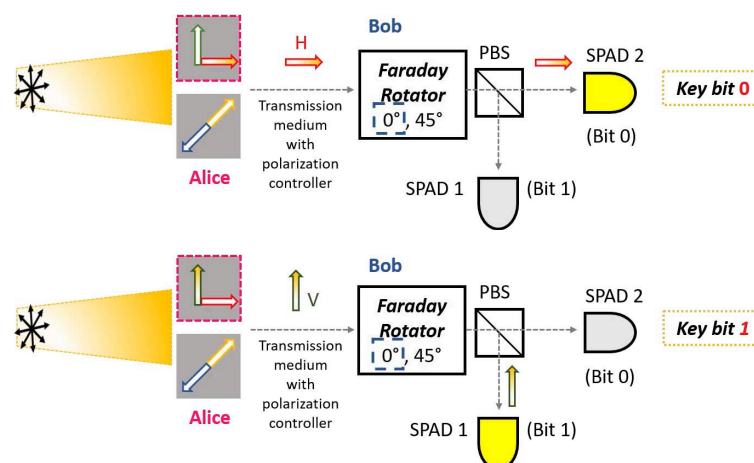


Figure 1. BB84 implementation using two SPADs.

We propose a novel BB84 implementation using a single SPAD, as shown in Fig. 2. In order to discriminate bit 0 and 1, Bob has to perform a further binary choice (which remains secret, i.e. not communicated to Alice through the public channel used for the sharing of basis choice between Alice and Bob). This choice is implemented as a Faraday rotation of 0° or 90° , which adds to the rotation of either 0° or 45° corresponding to the Bob's basis choice. Finally, a sifted key bit is obtained when Alice and Bob choose the same basis and there is a photon detection in the SPAD. Then the bit value 1 or 0 depends on the added rotation 0° or 90° , chosen by Bob in addition to the rotation representing the basis choice.

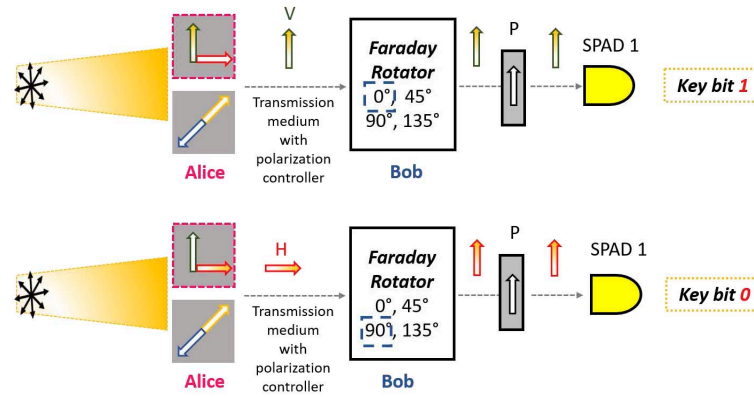


Figure 2. BB84 implementation using a single SPAD.

3. EXPERIMENTAL RESULTS AND CONCLUSIONS

We have realized the single-SPAD implementation of BB84 described in the above section considering as source of photons a strongly attenuated semiconductor diode laser emitting at the wavelength of 1553 nm. The photon polarization is controlled by Alice, setting one linear state among four states (i.e., horizontal, vertical, diagonal and antidiagonal) through a Polarization State Generator (PSG) device. After some meters of fiber propagation a manually-adjustable fiber polarization controller is used to compensate for the polarization change and then the beam is collimated in free-space, crossing a Variable Faraday Rotator (VFR) and a vertical analyzer. Finally, the detection of the single photons is carried out by Bob through an InGaAs/InP SPAD described in [3] and produced by MPD. The SPAD is set in gated mode and synchronized with PSG and VFR. In Fig. 3 the measured values of quantum bit-error rate (QBER) are reported for different values of the SPAD gate window width, between 10 and 500 ns. The experimentation carried out allows for obtaining a sifted key with a QBER of 1.3%, which is well below the accepted QBER limit (11%) for secure QKD [4].

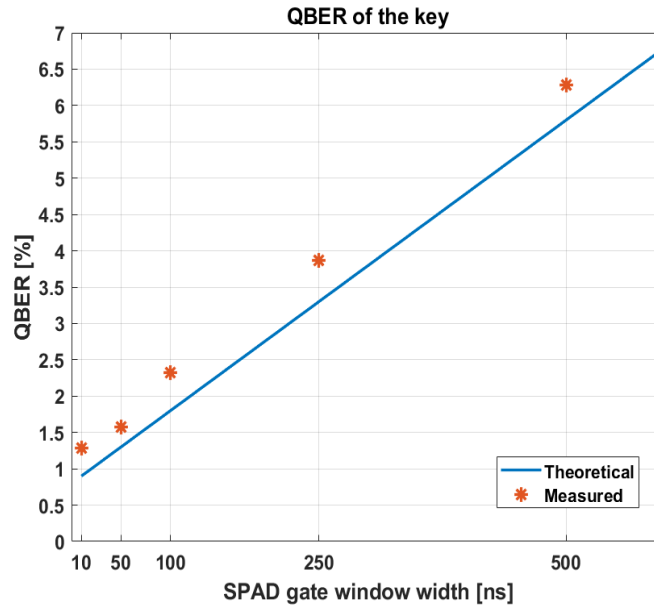


Figure 3. Measured QBER as a function of the SPAD gate window width.

4. CONCLUSIONS

The feasibility of a novel BB84 implementation has been experimentally proved, where a single SPAD, in combination with a VFR giving a polarization rotation chosen among four possible values (0°, 45°, 90° and 135°), is used by Bob. This single-SPAD solution presents a reduced cost with respect to the usual BB84 implementation with a pair of SPADs, but at the expense of a halved rate of generation of the key, because the cases in which the photon is blocked by the polarization analyzer are discarded.

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